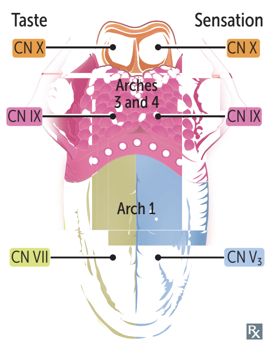
▸ GASTROINTESTINAL—EMBRYOLOGY

Tongue development



1st pharyngeal arch forms anterior 2/3 of tongue (sensation via CN V_3, taste via CN VII).

3rd and 4th pharyngeal arches form posterior 1/3 of tongue (sensation and taste mainly via CN IX, extreme posterior via CN X).

Motor innervation is via CN XII to hyoglossus (retracts and depresses tongue), **genio**glossus (**protrudes** tongue), and **styl**oglossus (draws sides of tongue upward to create a trough for swallowing).

Motor innervation is via CN X to palatoglossus (elevates posterior tongue during swallowing).

Taste—CN VII, IX, X (nucleus tractus solitarius [NTS]).

Pain—CN V_3, IX, X.

Motor—CN X, XII.

The **genie** comes **out** of the lamp in **styl**e.

CN **10** innervates pala**ten**glossus.

Normal gastrointestinal embryology

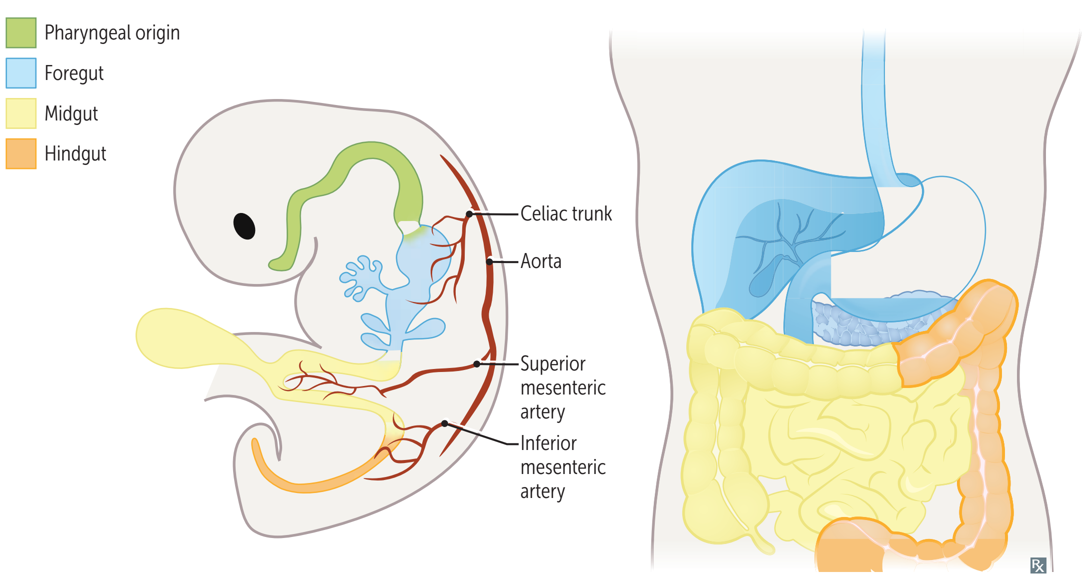
Foregut—esophagus to duodenum at level of pancreatic duct and common bile duct insertion (ampulla of Vater).

- 4th-6th week of development—stomach rotates 90° clockwise.
- Left vagus becomes anteriorly positioned, and right vagus becomes posteriorly positioned.

Midgut—lower duodenum to proximal 2/3 of transverse colon.

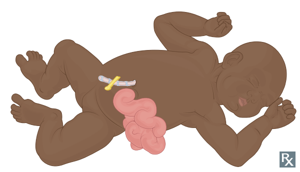
- 6th week of development—physiologic herniation of midgut through the umbilical ring. Yolk sac and midgut are connected via the vitelline (omphalomesenteric) duct.
- 10th week of development—returns to abdominal cavity rotating around superior mesenteric artery (SMA), 270° counterclockwise (~90° before 10th week, remaining ~180° in 10th week when contents retract back into abdominal cavity).

Hindgut—distal 1/3 of transverse colon to anal canal above pectinate line.



Ventral wall defects

Developmental defects due to failure of rostral fold closure (eg, sternal defects [ectopia cordis]), lateral fold closure (eg, omphalocele, gastroschisis), or caudal fold closure (eg, bladder exstrophy).

	Gastroschisis	Omphalocele
PRESENTATION	Paraumbilical herniation of abdominal contents through abdominal wall defect	Herniation of abdominal contents through umbilical ring
COVERAGE	Not covered by peritoneum or amnion A; right sided/paraumbilical	Covered by peritoneum and amnion B (light gray shiny sac); midline, membrane covered
ASSOCIATIONS	Not commonly associated with chromosomal abnormalities; good prognosis	Associated with congenital anomalies (eg, trisomies 13 and 18, Beckwith-Wiedemann syndrome) and other structural abnormalities (eg, cardiac, GU, neural tube)
	 	 

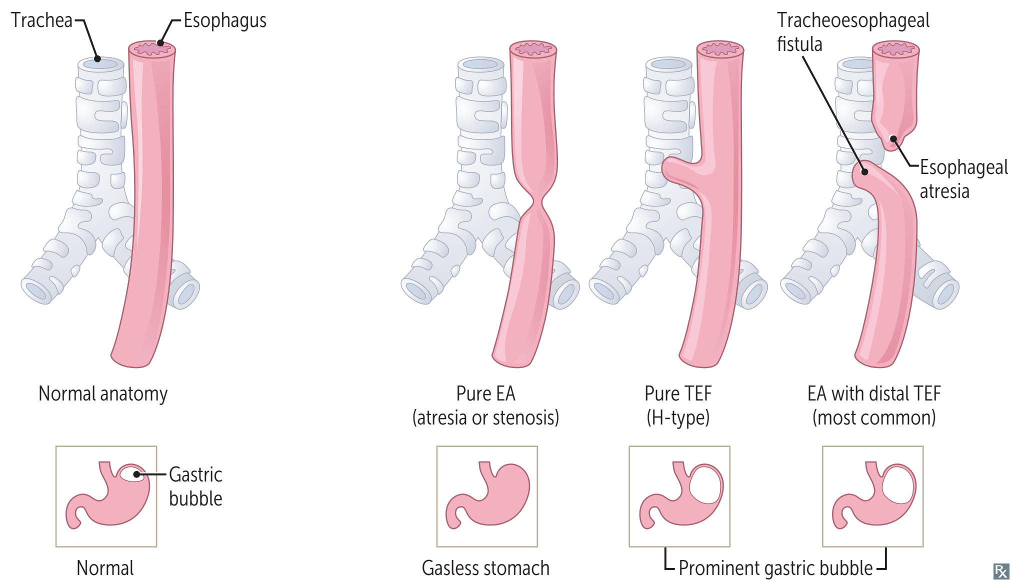
Congenital umbilical hernia

Delay of umbilical ring to close spontaneously following physiological herniation of midgut → patent umbilical orifice. Covered by skin C. Often reducible, but protrudes with ↑ intra-abdominal pressure (eg, crying). May be associated with congenital disorders (eg, Down syndrome, congenital hypothyroidism). Small defects usually close spontaneously.

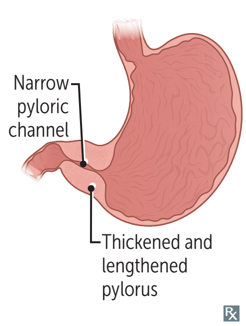
Tracheoesophageal anomalies

Esophageal atresia (EA) with distal tracheoesophageal fistula (TEF) is the most common (85%) and often presents as polyhydramnios in utero (due to inability of fetus to swallow amniotic fluid). Neonates drool, choke, and vomit with first feeding. TEFs allow excess air to enter stomach (visible on CXR as a prominent gastric bubble). Cyanosis is 2° to laryngospasm (to avoid reflux-related aspiration). Clinical test: failure to pass nasogastric tube into stomach. Associated with VATER/VACTERL defects.

In **H**-type, the fistula resembles the letter **H**. In pure EA, CXR shows gasless abdomen.



Hypertrophic pyloric stenosis



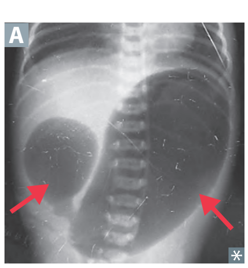
Most common cause of gastric outlet obstruction in infants. Palpable olive-shaped **m**ass (due to hypertrophy and hyperplasia of pyloric sphincter muscle) in epigastric region, visible peristaltic waves, and postprandial nonbilious projectile vomiting at ~ 2–6 weeks old. More common in firstborn **m**ales; associated with exposure to **m**acrolides.

Results in hypokalemic hypochloremic **m**etabolic alkalosis (2° to vomiting of gastric acid and subsequent volume contraction).

Ultrasound shows thickened and lengthened pylorus.

Treatment: surgical incision of pyloric muscles (pyloro**m**yotomy).

Intestinal atresia

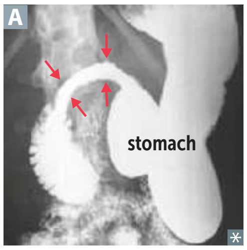


Presents with bilious vomiting and abdominal distension within first 1–2 days of life.

Duodenal atresia—failure to recanalize lumen from solid cord stage. X-ray A shows "**d**ouble bubble" (dilated stomach, proximal duodenum). Associated with **D**own syndrome.

Jejunal and ileal atresia—disruption of mesenteric vessels (typically SMA) → ischemic necrosis of fetal intestine → segmental resorption: bowel becomes discontinuous. X-ray may show "triple bubble" (dilated stomach, duodenum, proximal jejunum) and gasless colon. Associated with cystic fibrosis and gastroschisis. May be caused by maternal tobacco smoking or use of vasoconstrictive drugs (eg, cocaine) during pregnancy.

Pancreas and spleen embryology

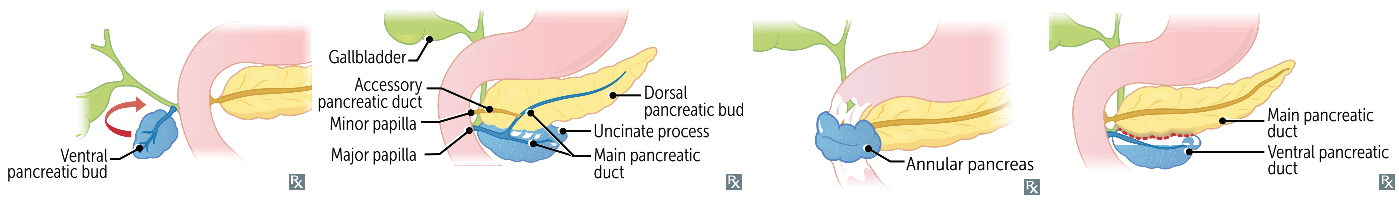


Pancreas—derived from foregut. Ventral pancreatic bud contributes to uncinate process. Both ventral and dorsal buds contribute to pancreatic head and main pancreatic duct.

Annular pancreas—abnormal rotation of ventral pancreatic bud forms a ring of pancreatic tissue → encircles 2nd part of duodenum; may cause duodenal narrowing (arrows in A) and vomiting. Associated with Down syndrome.

Pancreas divisum—ventral and dorsal parts fail to fuse at 7 weeks of development. Common anomaly; mostly asymptomatic, but may cause chronic abdominal pain and/or pancreatitis.

Spleen—arises in mesentery of the stomach (dorsal mesogastrium, hence, mesodermal), but has foregut supply (celiac trunk → splenic artery).



▶ GASTROINTESTINAL—ANATOMY

Retroperitoneal structures

Retroperitoneal structures A are posterior to (and outside of) the peritoneal cavity. Injuries to retroperitoneal structures can cause blood or gas accumulation in retroperitoneal space.

SAD PUCKER:
- Suprarenal (adrenal) glands [not shown]
- Aorta and IVC
- Duodenum (2nd through 4th parts)
- Pancreas (except tail)
- Ureters [not shown]
- Colon (descending and ascending)
- Kidneys
- Esophagus (thoracic portion) [not shown]
- Rectum (partially) [not shown]